

Performance Testing

Date	1 July 2026
Project Name	OptiCrop
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the OptiCrop system behaves under expected and peak load conditions when farmers, researchers, and policymakers submit soil and environmental data for crop recommendations. The sections below document the testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter (with Locust used for supplementary spike-test validation)
Type of Testing	Load Testing, Stress Testing, and Spike Testing
Target Module / API	/predict-crop endpoint of the Smart Crop Recommendation module (accepts N, P, K, temperature, humidity, pH, and rainfall values and returns the recommended crop)
Test Environment	Cloud – Flask/Python ML API hosted on an AWS EC2 instance behind an Nginx reverse proxy
Test Date	15 March 2026

Step 2: Test Scenarios

S. No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Farmers submit N, P, K, temperature, humidity, pH, and rainfall values via the Smart Crop Recommendation form (Scenario 1) to get the best-suited crop.	50	60	API returns an accurate crop recommendation within acceptable response time under normal load.
2	Users query the Crop Suitability and Environmental Assessment module (Scenario 2) to check whether current soil/climate conditions suit a specific crop.	100	120	System returns correct suitability and productivity-potential results with no errors under moderate concurrent load.
3	Agricultural researchers and policymakers run bulk analytical queries (Scenario 3) to	200	180	System processes bulk/analytical

	study crop-environment patterns for policy planning.			requests concurrently without degradation in accuracy or excessive delay.
4	Spike test simulating a sudden surge of farmers accessing the recommendation engine during peak sowing season.	500	60	System remains stable under sudden peak load with no crashes and graceful recovery after the spike.

Step 3: Performance Test Results

S. No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 seconds	Pass	Well within target under normal load.
2	Response Time (Max)	< 5 seconds	3.81 seconds	Pass	Peak observed during the 500-user spike test.
3	Throughput (Req/sec)	> 100 req/sec	128 req/sec	Pass	Measured during the 200-VU load test scenario.
4	Error Rate	< 1%	0.32%	Pass	Occasional timeouts only during spike test.
5	CPU Utilization	< 80%	68%	Pass	Peaked during bulk research-query scenario.
6	Memory Utilization	< 80%	72%	Pass	ML model kept in memory across requests; stable throughout.

Step 4: Observations & Analysis

Key Findings:

The OptiCrop recommendation engine maintained response times well under the 2-second target for average load and stayed under the 5-second ceiling even during the peak spike scenario. Throughput scaled predictably as virtual users increased from 50 to 200, and the system continued serving crop recommendations, suitability assessments, and research queries accurately across all three business scenarios without noticeable accuracy degradation.

Bottlenecks Identified:

- Slight latency increase was observed in the Crop Suitability and Environmental Assessment module (Scenario 2) when concurrent users exceeded 100, traced to synchronous database lookups for historical crop-environment data.
- During the 500-user spike test, brief request queuing occurred at the ML model inference layer, causing the small rise in maximum response time and error rate.

Optimization Steps Taken:

- Introduced database connection pooling and query caching for repeated environmental parameter lookups used in Scenario 2 and Scenario 3.
- Pre-loaded and kept the trained ML model resident in memory (instead of reloading per request) to reduce inference overhead during traffic spikes.
- Added request-level rate limiting and horizontal auto-scaling on the cloud instance to smooth out sudden surges in farmer traffic during peak sowing season.